

IntraPortal – Web Exploitation Lab Report

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Topic: SQL Injection, File Upload Exploitation & Privilege Escalation

Overview

This assessment focused on identifying weaknesses inside the IntraPortal lab environment. The exploitation process demonstrated how several security flaws could be chained together to move from a simple web vulnerability to full system compromise.

SQL Injection Authentication Bypass

The login system failed to properly sanitize user-controlled input before sending it to the database. By injecting a crafted SQL condition into the username field, authentication checks were bypassed and administrative access to the dashboard was obtained. This confirmed insecure query construction and missing parameterized statements.

Sensitive Information Exposure

After gaining administrative access, the configuration panel exposed operational details in plaintext. The page revealed internal account credentials, upload directory paths, and application configuration data. Exposed credentials significantly increased the impact of the initial compromise because they allowed direct access to backend services.

File Upload Validation Weakness

The upload functionality attempted to restrict users to image files only, but the validation process relied mainly on MIME types and filename filtering. By disguising a malicious payload as an image and modifying request metadata, it was possible to bypass upload restrictions and place a server-interpreted file inside a web-accessible directory.

Remote Code Execution

Once the uploaded file became reachable through the browser, the server executed it under the web service account context. This provided command execution capabilities on the operating system and confirmed Remote Code Execution on the target machine.

Privilege Escalation

Further enumeration showed that Python could be executed with elevated privileges through sudo without password verification. This misconfiguration allowed escalation from a low-privileged shell to full root access on the system.

Security Impact

The attack chain demonstrated how multiple vulnerabilities can combine into a complete infrastructure compromise. The weaknesses allowed unauthorized authentication, sensitive data exposure, command execution, and administrative control of the server.

Findings Summary

ID	Finding	Severity	Impact
F1	SQL Injection in Login System	Critical	Administrative access
F2	Plaintext Credential Exposure	High	Direct SSH access
F3	Improper File Upload Validation	Critical	Remote Code Execution
F4	Unsafe sudo Configuration	Critical	Root privilege escalation

Conclusion

The IntraPortal lab highlighted the importance of secure authentication logic, strict file upload controls, proper credential handling, and least-privilege system configuration. A single exploitable weakness at the web layer ultimately resulted in full root-level compromise of the host.